

MARMARA UNIVERSITY
FACULTY OF ENGINEERING



EE4077

FUNDAMENTALS OF MACHINE LEARNING

HOMEWORK II
SOLUTIONS

SOLUTION FOR QUESTION 1

a)

RSS for rabbits and mice experiment groups can be written as follows

$$RSS(w_R) = \sum_n \left[y_{R_n} - {x_{R_n}}^T w_R\right]^2$$

$$RSS(w_M) = \sum_n \left[y_{M_n} - {x_{M_n}}^T w_M\right]^2$$

b)

Solution for total RSS can be calculated as follows

$$RSS(\widehat{w}) = RSS(w_R) + RSS(w_R) = \sum_n \left[y_{R_n} - {x_{R_n}}^T \widehat{w}\right]^2 + \sum_n \left[y_{M_n} - {x_{M_n}}^T \widehat{w}\right]^2$$

$$\nabla RSS(\widehat{w}) = 2{X_R}^T X_R \widehat{w} - 2{X_R}^T y_R + 2{X_M}^T X_M \widehat{w} - 2{X_M}^T y_M = 0$$

$${X_R}^T X_R \widehat{w} - {X_R}^T y_R + {X_M}^T X_M \widehat{w} - {X_M}^T y_M = 0$$

$$\left({X_R}^T X_R + {X_M}^T X_M\right)\widehat{w} = {X_M}^T y_M + {X_R}^T y_R$$

$$\widehat{w} = \left({X_R}^T X_R + {X_M}^T X_M\right)^{-1}\left({X_M}^T y_M + {X_R}^T y_R\right)$$

c)

$\widehat{w}_R$and $\widehat{w}_M$ parameters that minimizes the regularized loss function can be calculated as follows

$$\widehat{w}_R = \left({X_R}^T X_R + \lambda I\right)^{-1} {X_R}^T y_R$$

$$\widehat{w}_M = \left({X_M}^T X_M + \lambda I\right)^{-1} {X_M}^T y_M$$

d)

As long as parameter λ is increased, $\|\widehat{w}_R - \widehat{w}_M\|$ will decrease since λ increases regularized term will be more dominant and effect the behavior of loss function.

e)

As λ goes infinity, the term $\frac{\|\widehat{w}_R\|}{\widehat{w}}$ and $\frac{\|\widehat{w}_M\|}{\widehat{w}}$ will yield to one since at infinity vector that minimizes total RSS and individual RSS values will be the same.

SOLUTION FOR QUESTION 2

a)

MLE estimate of p can be calculated as follows

$$\hat{\theta}_{mle} = \frac{n_H}{n_H + n_T} = \frac{490}{490 + 510} = 0.49$$

SOLUTION FOR QUESTION 3

3.1)

Given 3 training points can be written in the following equation sets

$$2w_1 - w_2 = 3$$

$$4w_1 - 2w_2 = 1$$

$$6w_1 - 3w_2 = 7$$

To solve this system of equation, following matrix notation can be used

$$w = (X^T X)^{-1} X^T y$$

where

$$X = \begin{bmatrix} 2 & -1 \\ 4 & -2 \\ 6 & -3 \end{bmatrix} \; and \; y = \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}$$

$$w = \left(\begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 4 & -2 \\ 6 & -3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}$$

$$w = \left(\begin{bmatrix} 56 & -28 \\ -28 & 56 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}$$

This has no solution since matrix $X^T X$ has no inverse. To find a solution for this set of equation (dataset) we can use regularization. Then w can be computed as follows

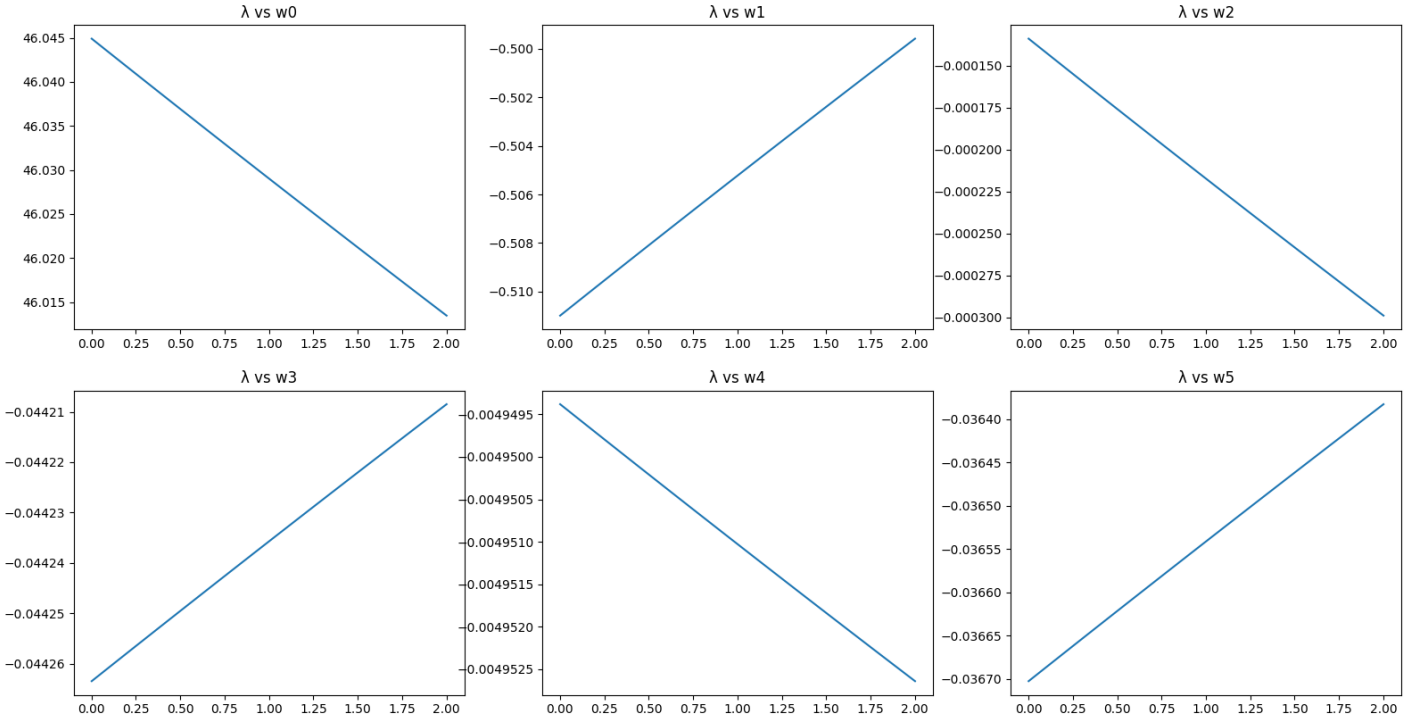
$$w = (X^T X + \lambda I)^{-1} X^T y$$

Then, for λ = 5

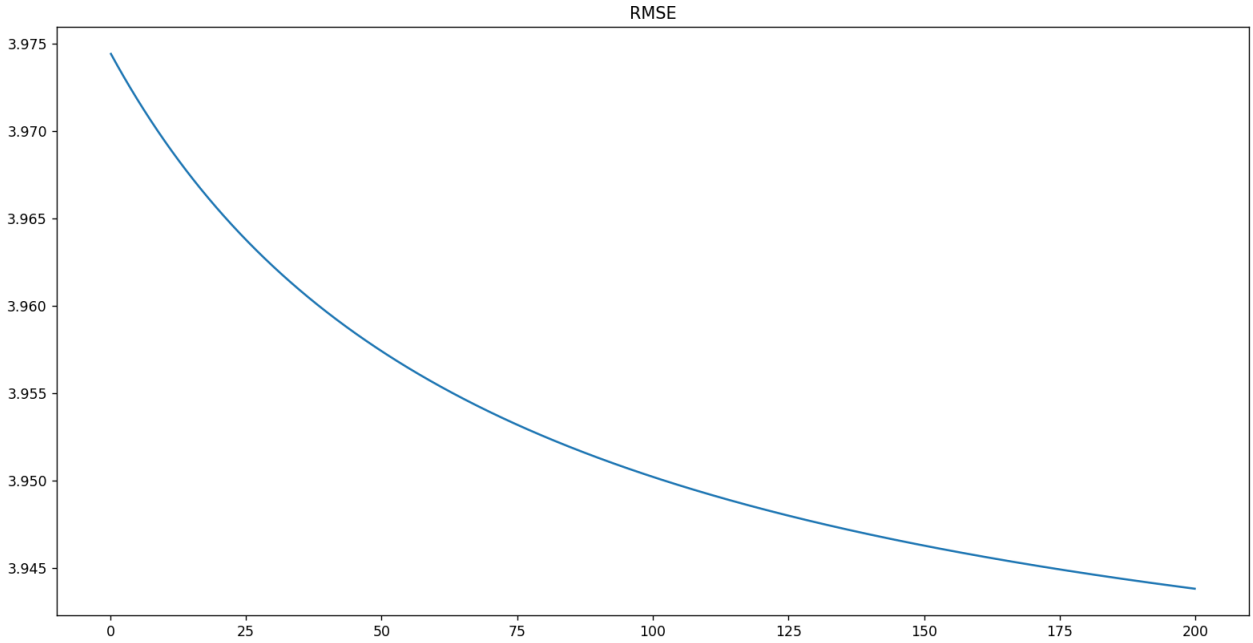
$$w = \left(\begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 4 & -2 \\ 6 & -3 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 & 4 & 6 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}$$

$$w = \begin{bmatrix} 0.6933 \\ -0.3467 \end{bmatrix}$$

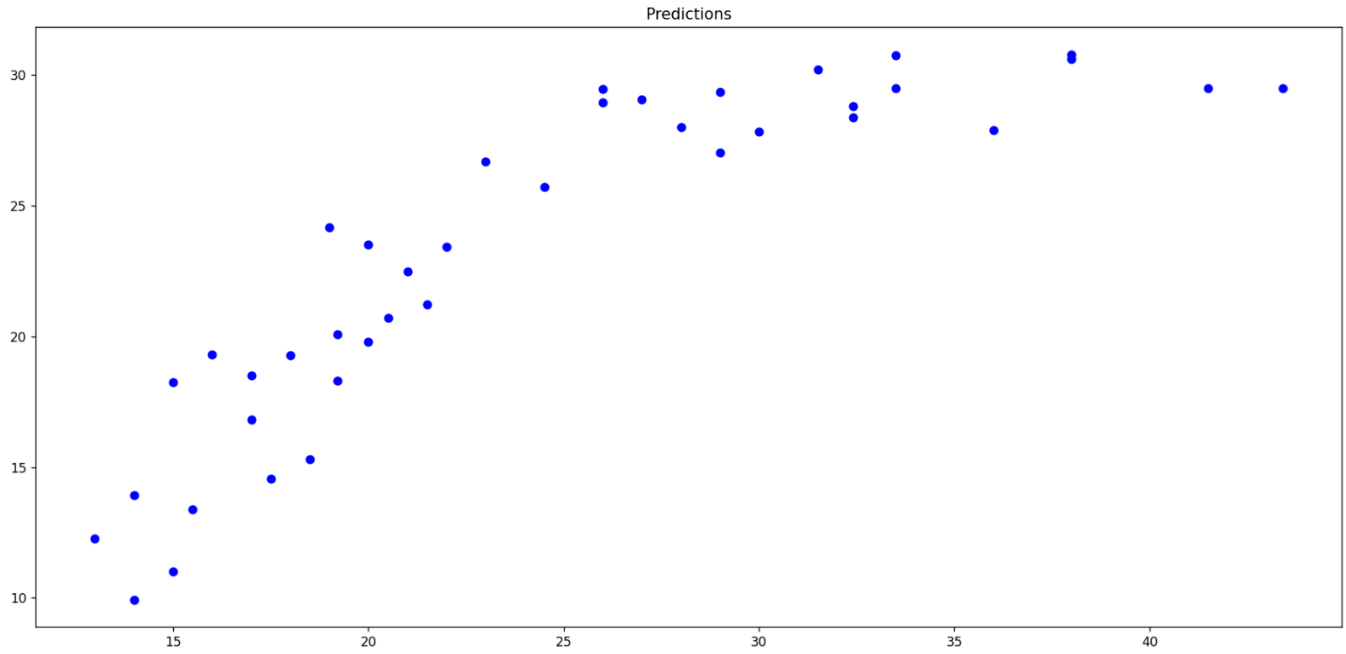
3.3)



As long as parameter λ is increased, coefficients and intercept is starting to converge zero. Some of them will be very close to zero when λ is going to infinity which make our model overfit.



As λ increases, the RMSE score is decreased. The value λ = 100 is selected according RMSE plot.



The code for question 3 can be found in Appendix.A section of this PDF.

SOLUTION FOR QUESTION 4

$$Objective\ 3: \min_{\boldsymbol{w}} \sum_{i=1}^{n} \left(y_i - \boldsymbol{x}_i^T \boldsymbol{w}\right)^2$$

$$Objective\ 4: \min_{\boldsymbol{w}} \sum_{i=1}^{n} \left(y_i - \boldsymbol{x}_i^T \boldsymbol{w}\right)^2 + \lambda \sum_{j=1}^{8} (w_j)^2$$

$$Objective\ 3: \min_{\boldsymbol{w}} \sum_{i=1}^{n} \left(y_i - \boldsymbol{x}_i^T \boldsymbol{w}\right)^2 + \lambda \sum_{j=1}^{8} |w_j|$$

a)

Objective 4 and Objective 5 has regularization term. Objective 4 has L2 norm regularization where Objective 3 has L1 norm regularization.

b)

As long as λ increases, the bias would also increase since regularization term is dominates minimizing the RSS which makes model simpler. It avoiding model to overfit at the cost of underfit.

c)

Similarly, as long as λ increases variance of the model may increase since some of the model parameters are tend to zero which means simple model higher variance.

d)

Objective 3 → Col C → Seems no regularization since lots of values are zero and every other value is positive.

Objective 4 → Col A → Seems L2 regularization since less value is near zero and negative and positive coefficients are together.

Objective 5 → Col B → Seems L1 regularization since more value is near zero and negative and positive coefficients are together.

SOLUTION FOR QUESTION 5

a)

Number of occurance for each event X is shown in the below table

$X_1 = 1$	$X_1 = 0$	$X_2 = 1$	$X_2 = 0$	$X_3 = 1$	$X_3 = 0$	Y	$Total$
2	1	1	2	2	1	0	3
2	1	1	2	1	2	1	3

Probability of $P(Y = 0 \,|X_1 = 0, X_2 = 1, X_3 = 1)$ can be calculated as follows

$$P(Y = 0 \,|X_1 = 0, X_2 = 1, X_3 = 1) = P(Y = 0)P(X_1 = 0|Y = 0)P(X_2 = 1|Y = 0)P(X_3 = 1|Y = 0)$$

$$P(Y = 0 \,|X_1 = 0, X_2 = 1, X_3 = 1) = \frac{3}{6}\ \frac{1}{3}\ \frac{1}{3}\ \frac{2}{3} = \frac{1}{27}$$

b)

After applying Laplace smoothing with k = 1, above table changes as follows

$X_1 = 1$	$X_1 = 0$	$X_2 = 1$	$X_2 = 0$	$X_3 = 1$	$X_3 = 0$	Y	$Total$
3	2	2	3	3	2	0	5
3	2	2	3	2	3	1	5

Probability of $P(Y = 1 \,|X_1 = 0, X_2 = 1, X_3 = 0)$ can be calculated as follows

$$P(Y = 1 \,|X_1 = 0, X_2 = 1, X_3 = 0) = P(Y = 1)P(X_1 = 0|Y = 1)P(X_2 = 1|Y = 1)P(X_3 = 0|Y = 1)$$

$$P(Y = 1 \,|X_1 = 0, X_2 = 1, X_3 = 0) = \frac{5}{10}\ \frac{2}{5}\ \frac{2}{5}\ \frac{3}{5} = \frac{6}{125}$$

c)

After applying Laplace smoothing with k = 5, above table changes as follows

$X_1 = 1$	$X_1 = 0$	$X_2 = 1$	$X_2 = 0$	$X_3 = 1$	$X_3 = 0$	Y	$Total$
7	6	6	7	7	6	0	13
7	6	6	7	6	7	1	13

Probability of $P(Y = 1 \,|X_1 = 0, X_2 = 1, X_3 = 0)$ can be calculated as follows

$$P(Y = 1 \,|X_1 = 0, X_2 = 1, X_3 = 0) = P(Y = 1)P(X_1 = 0|Y = 1)P(X_2 = 1|Y = 1)P(X_3 = 0|Y = 1)$$

$$P(Y = 1 \,|X_1 = 0, X_2 = 1, X_3 = 0) = \frac{13}{26}\ \frac{6}{13}\ \frac{6}{13}\ \frac{7}{13} = \frac{126}{2197}$$

Probability is decreased, since we added new terms for every possible input for X input vector with Laplace smoothing.

SOLUTION FOR QUESTION 6

6.1)

a)

Basis function for the given model is

$$\varphi_k(x) = x^k$$

b)

RSS in terms of $\varphi_k(x)$ can be calculated as follows

$$RSS(w) = \sum_n (w^T \varphi_n(x) - y_n)^2$$

c)

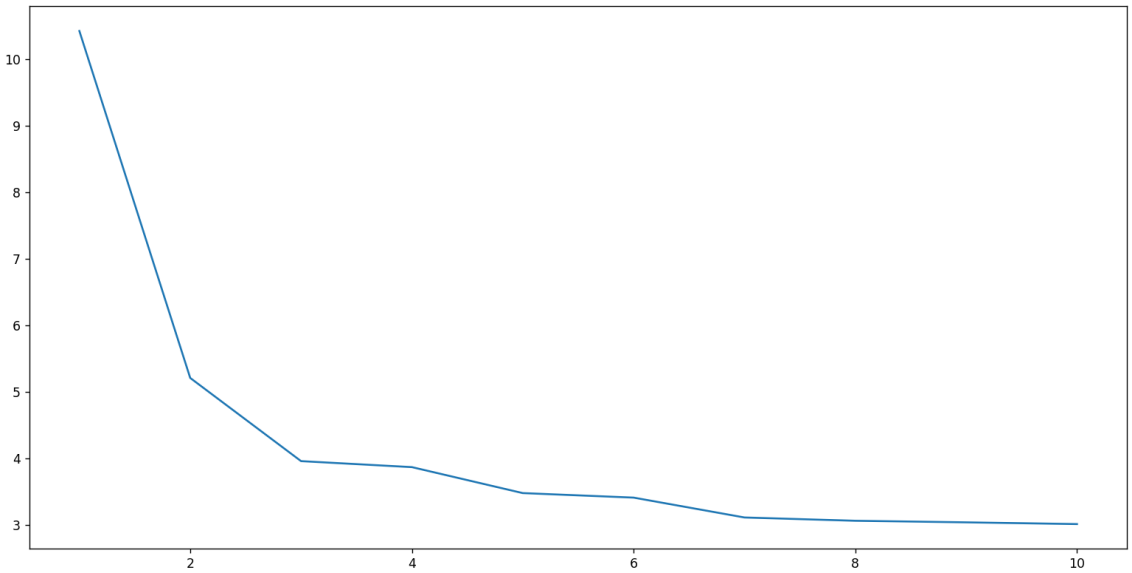
w that minimizes RSS can be calculated as follows

$$w = \left(\Phi^T \Phi\right)^{-1} \Phi^T y$$

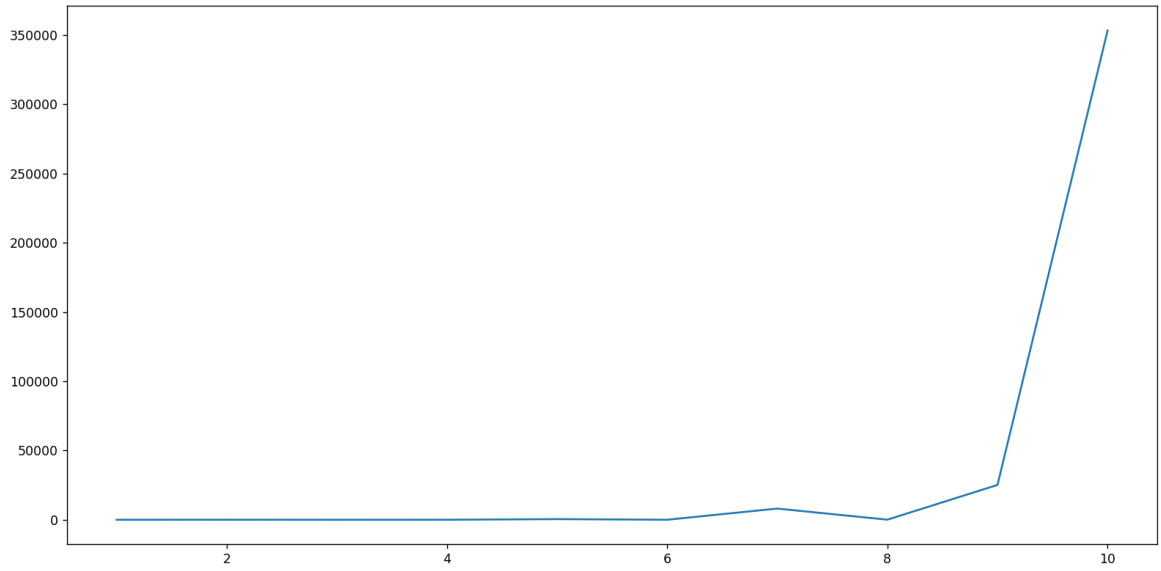
where

$$\Phi = \begin{bmatrix} \varphi(x_1)^T \\ \varphi(x_2)^T \\ \vdots \\ \varphi(x_N)^T \end{bmatrix}$$

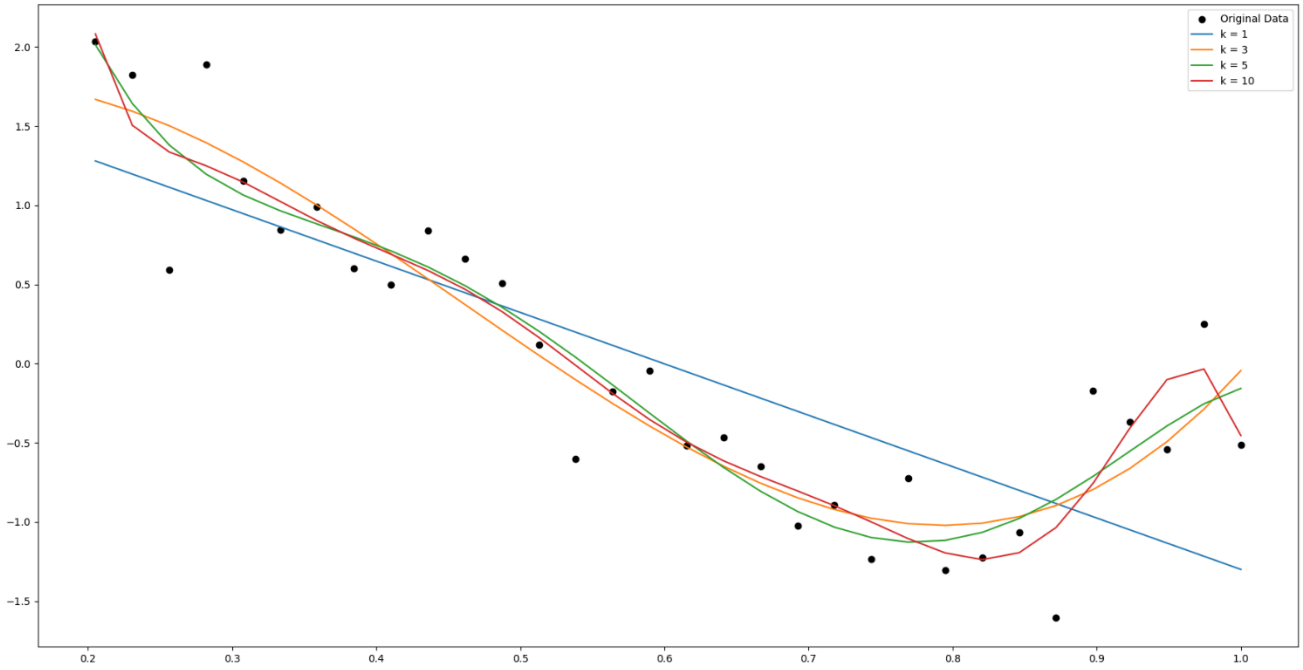
6.3)



According to training set, best RSS is computed for k = 10.



According to validation set, best RSS is computed for k = 6. The reason of parameter k different in training and validation set is, during training increasing k means model is overfit to data and starting to memorize. For that training set it will be the best model but future inputs it yields much more error.



The code for question 6 can be found in Appendix.B section of this PDF.

APPENDIX.A Code for Question 3

```
import pandas as pd
import numpy as np
from sklearn.linear_model import Ridge
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt

def fit(X, y, a):
    model = Ridge(alpha=a)
    model.fit(X,y)

    return model

def predict(X, w):
    y_pred = w.predict(X)

    return y_pred


df = pd.read_csv("car.csv")
df = df.drop(["name","origin","year"], axis='columns')
X = df.drop("mpg", axis='columns')
Y = df["mpg"]

X_, X_test, Y_, Y_test = train_test_split(X, Y, test_size=0.10,
random_state=1)
X_train, X_valid, Y_train, Y_valid = train_test_split(X_, Y_,
test_size=0.113, random_state=1)

a = np.arange(0.001, 2, 0.002)

w0 = []
w1 = []
w2 = []
w3 = []
w4 = []
w5 = []
rmse = []

for alpha in a:
    model = fit(X_train, Y_train, alpha)
    w0.append(model.intercept_)
    w1.append(model.coef_[0])
    w2.append(model.coef_[1])
    w3.append(model.coef_[2])
    w4.append(model.coef_[3])
    w5.append(model.coef_[4])
    Y_pred = predict(X_valid, model)
    rmse.append(mean_squared_error(Y_valid, Y_pred, squared=False))


#Best lambda for lower RMSE
model = fit(X_train, Y_train, 100)
Y_pred = predict(X_test, model)


#Figure 1
plt.figure(1)
```

```
# Subplot 1
plt.subplot(2, 3, 1)
plt.plot(a, w0)
plt.title('λ vs w0')

# Subplot 2
plt.subplot(2, 3, 2)
plt.plot(a, w1)
plt.title('λ vs w1')

# Subplot 3
plt.subplot(2, 3, 3)
plt.plot(a, w2)
plt.title('λ vs w2')

# Subplot 4
plt.subplot(2, 3, 4)
plt.plot(a, w3)
plt.title('λ vs w3')

# Subplot 5
plt.subplot(2, 3, 5)
plt.plot(a, w4)
plt.title('λ vs w4')

# Subplot 6
plt.subplot(2, 3, 6)
plt.plot(a, w5)
plt.title('λ vs w5')

#Figure 2
plt.figure(2)
plt.plot(a, rmse)
plt.title("RMSE")

#Figure 3
plt.figure(3)
plt.scatter(Y_test, Y_pred, marker='o', color='b', label='Dots')
plt.title("Predictions")

# Adjust layout
plt.tight_layout()

# Show the plots
plt.show()
```

APPENDIX.B Code for Question 6

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

def fit(Xtraining_poly, Ytraining_poly, k):
    poly = PolynomialFeatures(degree=k, include_bias=False)
    #reshape data to work properly with sklearn
    poly_features = poly.fit_transform(Xtraining_poly.reshape(-1, 1))
    poly_reg_model = LinearRegression()
    poly_reg_model.fit(poly_features, Ytraining_poly)

    weights = []
    weights.append(poly_reg_model.intercept_)
    weights.extend(poly_reg_model.coef_)

    return weights

def validate(Xvalidation_poly, Yvalidation_poly, w):
    RSS = 0
    y_predictions =[]
    for idx, val in enumerate(Xvalidation_poly):
        y_pred = 0
        for i in range(0, len(w)):
            y_pred = y_pred + ((val ** i) * w[i])
        y_predictions.append(y_pred)
        RSS = RSS + ((Yvalidation_poly[idx] - y_pred) ** 2)

    return RSS,y_predictions



X_training =
pd.read_csv("Xtraining_poly.csv").to_numpy(dtype='float').transpose()[0]
X_validation =
pd.read_csv("Xvalidation_poly.csv").to_numpy(dtype='float').transpose()[0]
Y_training =
pd.read_csv("Ytraining_poly.csv").to_numpy(dtype='float').transpose()[0]
Y_validation =
pd.read_csv("Yvalidation_poly.csv").to_numpy(dtype='float').transpose()[0]


degree = np.arange(1, 11, 1)
RSS_train = []
RSS_valid = []
for k in degree:
    w = fit(X_training, Y_training, k)
    rss_train, _ = validate(X_training, Y_training, w)
    rss_valid, _ = validate(X_validation, Y_validation, w)
    RSS_train.append(rss_train)
    RSS_valid.append(rss_valid)

plt.figure(1)
plt.plot(degree, RSS_train)
plt.figure(2)
plt.plot(degree, RSS_valid)
plt.figure(3)
plt.scatter(X_training, Y_training, color='black')
```

```
#Just for plotting, make it in a for loop
w = fit(X_training, Y_training, 1)
_, predictions1 = validate(X_training, Y_training, w)
plt.plot(X_training, predictions1)
w = fit(X_training, Y_training, 3)
_, predictions3 = validate(X_training, Y_training, w)
plt.plot(X_training, predictions3)
w = fit(X_training, Y_training, 5)
_, predictions5 = validate(X_training, Y_training, w)
plt.plot(X_training, predictions5)
w = fit(X_training, Y_training, 10)
_, predictions10 = validate(X_training, Y_training, w)
plt.plot(X_training, predictions10)


plt.legend(['Original Data', 'k = 1', 'k = 3', 'k = 5', 'k = 10'])

plt.tight_layout()
plt.show()
```